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00001      ** DL Register Area - 120Word
00002      **
00003      ** DW00000 - DW00009 (PLS Use)
00004      ** DW00010 - DW00039 (Bit Use)
00005      ** DW00040 - DW00089 (Word Use)
00006      ** DW00090 - DW00119 (Timer)
00007      ****
00008
00009      "# Program Name      : 1Scribe
00010      "# Functionior       :
00011      "# Call Program     :
00012      "# Sub Program      :
00013      "# Func Program     :
00014
00015      VAR;
00016          " Scribe Chuck
00017 00000      LONG IDX_SCHY1 = 48;
00018 00001      LONG IDX_SCHY2 = 49;
00019          " Scribe Z
00020 00002      LONG IDX_CZ = 34;
00021          " Scribe Cutter
00022 00003      LONG IDX_CX = 32;
00023 00004      LONG IDX_CAMX = 33;
00024 00005      LONG Axis_POS_CAMX = 0;
00025          " Scribe Shift
00026 00006      LONG IDX_CS = 35;
00027 00007      LONG Addr = 0;
00028          " Scribe Speed
00029 00008      LONG Scribe_Speed_1st = 0;
00030 00009      LONG Scribe_Speed_2nd = 0;
00031 00010      LONG Scribe_Speed_3rd = 0;
00032 00011      LONG SC_PRT_Addr_L = 0;
00033      END_VAR;
00034
00035 00012      Addr = IDX_CS * 50;
00036
00037 00013      SC_PRT_Addr_L = (ML15414 - 1) * 100;
00038
00039          " Manual/Auto 구분
00040 00014      DB00020 = 0;
00041 00015      DB00021 = 0;
00042 00016      t
00043 00017          DB00022 = 0;
00044 00018          IF MB00004 == 1
00045 00019              DB00020 = 1;
00046 00020          ELSE;
00047 00021              IF MB225580 == 1
00048 00022                  DB00020 = 1;
00049 00023              IEND;
00050 00024              IF MW00051 == 52;
00051 00025                  DB00021 = 1;
00052 00026              t
00053 00027                  IEND;
00054 00028                  IF MB000800 | !MB000802 == 1;
00055 00029                      DB00022 = 1;
00056                      IEND;
00057                      " Scribe Executing Flag
00058 00030                      IF !MB000800 & MB000802 == 1;
00059 00031                          MB15420F = 1;
00060 00032                      IEND;
00061
00062                      " Cutter Select
00063                      " UPCZ - LOCZ
00064 00033                      IF !MB11420 & !MB11440 == 1;
00065 00034                          MB158800 = 1;
00066 00035                          MB158801 = 0;
00067 00036                          MB158802 = 1;
00068 00037                          MB158803 = 0;
00069 00038                      IEND;
00070                      " UPCZ - LORZ
00071 00039                      IF !MB11420 & MB11440 == 1;
00072 00040                          MB158800 = 1;
00073 00041                          MB158801 = 0;
00074 00042                          MB158802 = 0;
00075 00043                          MB158803 = 1;
00076 00044                      IEND;
00077                      " UPRZ - LOCZ
00078 00045                      IF MB11420 & !MB11440 == 1;
00079 00046                          MB158800 = 0;
00080 00047                          MB158801 = 1;
00081 00048                          MB158802 = 1;
00082 00049                          MB158803 = 0;
00083 00050                      IEND;
00084
00085 00051      DB000000 = 0;

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" Manual Data
 " Scribe Complete Position 1 Cycle Selec
 " Idling & Don't Cutter
 " AutoRunning
 " Recipe Data
 " Recipe Data
 " Recipe Data
 " Scribe Complete Position 1 Cycle Selec
 " Idling & Don't Cutter
 " Upper Cutter Z
 " Upper Roller Z
 " Lower Cutter Z
 " Lower Roller Z
 " Upper Cutter Z
 " Upper Roller Z
 " Lower Cutter Z
 " Lower Roller Z
 " 1st Cutting Cmt

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00086 00052      DB000001 = 0;          " 2nd Cutting Cmt
00087 00053      DB000002 = 0;          " 3rd Cutting Cmt
00088 00054      DB000010 = 0;          " 1st Set Flag (0:Use 1:NotUse)
00089 00055      DB000011 = 0;          " 3rd Set Flag (0:Use 1:NotUse)
00090
00091 00056      PLD [CX][CS][CZ][CAMX];
00092
00093 00057      IF DB000020 == 0          " Manual Data
00094 00058          DB000010 = 1;          " 1st Set Flag (0:Use 1:NotUse)
00095 00059          DB000011 = 1;          " 3rd Set Flag (0:Use 1:NotUse)
00096 00060      ELSE;
00097 00061          IF ML15292 == 0;          " 1st Cutter X Speed(UPCX)
00098 00062              IF ML15294 == 0          " 1st Cutter X Speed(LOCX)
00099 00063                  DB000010 = 1;          " 1st Set Flag (0:Use 1:NotUse)
00100 00064                  IEND;
00101 00065              IEND;
00102 00066          IF ML15296 == 0;          " 3rd Cutter X Speed(UPCX)
00103 00067              IF ML15298 == 0          " 3rd Cutter X Speed(LOCX)
00104 00068                  DB000011 = 1;          " 3rd Set Flag (0:Use 1:NotUse)
00105 00069                  IEND;
00106 00070              IEND;
00107 00071          IF DB000021 == 1          " Scribe Complete Position 1 Cycle Selec

t
00108 00072              DB000010 = 1;          " 1st Set Flag (0:Use 1:NotUse)
00109 00073              DB000011 = 1;          " 3rd Set Flag (0:Use 1:NotUse)
00110 00074              IEND;
00111 00075          IEND;
00112
00113 00076      DL00100 = ML15210 - ML15202          " UPCX Cutting Distance
00114 00077      DL00102 = ML15212 - ML15204          " LOCX Cutting Distance
00115 00078      DL00104 = ML15222 - ML15218          " UPCS Cutting Distance
00116 00079      DL00106 = ML15224 - ML15220          " LOCS Cutting Distance
00117 00080      IF DL00100 < 0;
00118 00081          DL00100 = DL00100 * -1;
00119 00082      IEND;
00120 00083      IF DL00102 < 0;
00121 00084          DL00102 = DL00102 * -1;
00122 00085      IEND;
00123
00124      " 1st Cutting(Not) / 3rd Cutting(Not)
00125 00086      IF DB000010 & DB000011 == 1
00126          " 2nd
00127          " Upper Scribe End Position Cal
00128 00087      DL00310 = ML15210;
00129          " Lower Scribe End Position Cal
00130 00088      DL00312 = ML15212;
00131          " Shift Y End Position Calc
00132 00089      DL00314 = ML40030[Addr] + ML15220;
00133
00134          " Distance Calc
00135 00090      DL00200 = DL00310 - ML15202          " UPCX
00136 00091      DL00202 = DL00312 - ML15204          " LOCX
00137 00092      DL00204 = ML15222 - ML15218          " UPCS
00138 00093      DL00206 = ML15224 - ML15220          " LOCS
00139
00140          " 1st
00141 00094      DL00300 = DL00310;
00142 00095      DL00302 = DL00312;
00143          " 3rd
00144 00096      DL00320 = DL00310;
00145 00097      DL00322 = DL00312;
00146 00098      ELSE;
00147          " 1st Cutting(Not) / 3rd Cutting(Use)
00148 00099      IF DB000010 == 1;
00149          " 2nd
00150          " Upper Scribe End Position Cal
00151 00100      DL00310 = ML15210 - ML15288;
00152          " Lower Scribe End Position Cal
00153 00101      DL00312 = ML15212 - ML15290;
00154          " Shift Y End Position Calc
00155 00102      DF00424 = (DL00100 - ML15288) / DL00100 * DL00104;
00156 00103      DF00426 = (DL00102 - ML15290) / DL00102 * DL00106;
00157 00104      DL00434 = DF00424;
00158 00105      DL00436 = DF00426;
00159
00160 00106      DL00314 = ML40030[Addr] + ML15218 + DL0043
00161
00162          " 3rd
00163          " Upper Scribe End Position Cal
00164 00107      DL00320 = ML15210;
00165          " Lower Scribe End Position Cal
00166 00108      DL00322 = ML15212;
00167          " Shift Y End Position Calc
00168 00109      DL00324 = ML40030[Addr] + ML15222;
00169
00170          " Distance Calc
00171 00110      DL00200 = DL00310 - ML15202          " UPCX

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00172 00111      DL00202 = DL00312 - ML15204      " LOCX
00173 00112      DL00204 = DL00414                " UPCS
00174 00113      DL00206 = DL00416                " LOCS

00175
00176      " 1st
00177 00114      DL00300 = DL00310;
00178 00115      DL00302 = DL00312;
00179 00116      ELSE;
00180      " 1st Cutting(Use) / 3rd Cutting(Not)
00181 00117      IF DB000011 == 1
00182      " 1st
00183      " Upper Scribe End Position Cal
00184 00118      DL00300 = ML15202 + ML15284;
00185      " Lower Scribe End Position Cal
00186 00119      DL00302 = ML15204 + ML15286;
00187      " Shift Y End Position Calc
00188 00120      DF00404 = ML15284 / DL00100 * DL00104;
00189 00121      DF00406 = ML15286 / DL00102 * DL0010
00190 00122      DL00414 = DF00404;
00191 00123      DL00416 = DF00406;
00192
00193 00124      DL00304 = ML40030[Addr] + ML15218 + DL00414;
00194
00195      " 2nd
00196      " Upper Scribe End Position Cal
00197 00125      DL00310 = ML15210;
00198      " Lower Scribe End Position Cal
00199 00126      DL00312 = ML15212;
00200      " Shift Y End Position Calc
00201 00127      DL00314 = ML40030[Addr] + ML1522;
00202
00203      " Distance Calc
00204 00128      DL00200 = DL00310 - DL00300      " UPCX
00205 00129      DL00202 = DL00312 - DL00302      " LOCX
00206 00130      DL00204 = DL00314 - DL00304      " UPCS
00207 00131      DL00206 = DL00316 - DL00306      " LOCS
00208
00209      " 3rd
00210 00132      DL00320 = DL00310;
00211 00133      DL00322 = DL00312;
00212 00134      ELSE;
00213      " 1st Cutting(Use) / 3rd Cutting(Use)
00214      " 1st
00215      " Upper Scribe End Position Cal
00216 00135      DL00300 = ML15202 + ML15284;
00217      " Lower Scribe End Position Cal
00218 00136      DL00302 = ML15204 + ML15286;
00219      " Shift Y End Position Calc
00220 00137      DF00404 = ML15284 / DL00100 * DL00104;
00221 00138      DF00406 = ML15286 / DL00102 * DL0010
00222 00139      DL00414 = DF00404;
00223 00140      DL00416 = DF00406;
00224
00225 00141      DL00304 = ML40030[Addr] + ML15218 + DL0041
00226
00227      " 2nd
00228      " Upper Scribe End Position Cal
00229 00142      DL00310 = ML15210 - ML15288;
00230      " Lower Scribe End Position Cal
00231 00143      DL00312 = ML15212 - ML15290;
00232      " Shift Y End Position Calc
00233 00144      DF00424 = (DL00100 - ML15288) / DL00100 * DL00104;
00234 00145      DF00426 = (DL00102 - ML15290) / DL00102 * DL0010
00235 00146      DL00434 = DF00424;
00236 00147      DL00436 = DF00426;
00237
00238 00148      DL00314 = ML40030[Addr] + ML15218 + DL0043
00239
00240      " 3rd
00241      " Upper Scribe End Position Cal
00242 00149      DL00320 = ML15210;
00243      " Lower Scribe End Position Cal
00244 00150      DL00322 = ML15212;
00245      " Shift Y End Position Calc
00246 00151      DL00324 = ML40030[Addr] + ML1522;
00247
00248      " Distance Calc
00249 00152      DL00200 = DL00310 - DL00300      " UPCX
00250 00153      DL00202 = DL00312 - DL00302      " LOCX
00251 00154      DL00204 = DL00314 - DL00304      " UPCS
00252 00155      DL00206 = DL00316 - DL00306      " LOCS
00253 00156      IEND;
00254 00157      IEND;
00255 00158      IEND;
00256
00257      " Upper Cutter X

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00258 00159      IF ML01142 == 0                                " Upper Roller Use(0:Cutter 1:Roller)
00259 00160          DL00500 = DL00300
00260 00161          DL00510 = DL00310;
00261 00162          DL00520 = DL00320;
00262 00163      ELSE;
00263 00164          IF CW00001 == 1                                " Scribe Direction (0: Forward / 1: Reve
rse)
00264 00165              DL00500 = DL00300 + ML23746;
00265 00166              DL00510 = DL00310 + ML23746;
00266 00167              DL00520 = DL00320 + ML23746;
00267 00168          ELSE;
00268 00169              DL00500 = DL00300 - ML23746;
00269 00170              DL00510 = DL00310 - ML23746;
00270 00171              DL00520 = DL00320 - ML23746;
00271 00172          IEND;
00272 00173      IEND;
00273          " Lower Cutter X
00274 00174      IF ML01144 == 0                                " Lower Roller Use(0:Cutter 1:Roller)
00275 00175          DL00502 = DL00302
00276 00176          DL00512 = DL00312;
00277 00177          DL00522 = DL00322;
00278 00178      ELSE;
00279 00179          IF CW00001 == 1                                " Scribe Direction (0: Forward / 1: Reve
rse)
00280 00180              DL00502 = DL00302 + ML23748;
00281 00181              DL00512 = DL00312 + ML23748;
00282 00182              DL00522 = DL00322 + ML23748;
00283 00183          ELSE;
00284 00184              DL00502 = DL00302 - ML23748;
00285 00185              DL00512 = DL00312 - ML23748;
00286 00186              DL00522 = DL00322 - ML23748;
00287 00187          IEND;
00288 00188      IEND;
00289          " 스크라이브 재시작 동작 중
00290          DB000030 = 1;                                " 1st Cutting Enabl
00291 00189          DB000031 = 1;                                " 2nd Cutting Enabl
00292 00190          DB000032 = 1;                                " 3rd Cutting Enable
00293 00191          IF MB047350 == 1                            " Scribe Emergency Stop
00294 00192              IF CW00001 == 1                            " Scribe Direction (0: Forward / 1: Reve
rse)
00295 00193                  IF DL00500 > ML04910;
00296 00194                      DB000030 = 0;                                " 1st Cutting Enable
00297 00195                  IEND;
00298 00196                  IF DL00520 > ML04910;
00299 00197                      DB000031 = 0;                                " 2nd Cutting Enable
00300 00198                  IEND;
00301 00199                  ELSE;
00302 00200                      IF DL00500 < ML04910;
00303 00201                          DB000030 = 0;                                " 1st Cutting Enable
00304 00202                      IEND;
00305 00203                      IF DL00520 < ML04910;
00306 00204                          DB000031 = 0;                                " 2nd Cutting Enable
00307 00205                      IEND;
00308 00206                  IEND;
00309 00207                  IEND;
00310 00208          IEND;
00311          " Speed Calc and Settin
00312          DF00210 = DL00200 * DL00200                                " dx1
00313 00209          DF00212 = DL00202 * DL00202                                " dx2
00314 00210          DF00214 = DL00204 * DL00204                                " dy1
00315 00211          DF00216 = DL00206 * DL00206                                " dy2
00316 00212          " dx1^2 + dx2^2 + dy1^2 + dy2^2
00317          DF00220 = DF00210 + DF00212 + DF00214 DF00216;
00318          DF00222 = SQT(DF00220);
00319 00213          DF00224 = DF00210;
00320 00214          IF DF00224 < DF00212;
00321 00215              DF00224 = DF00212;
00322 00216          IEND;
00323 00217          IF DF00224 < DF00214;
00324 00218              DF00224 = DF00214;
00325 00219          IEND;
00326 00220          IF DF00224 < DF00216;
00327 00221              DF00224 = DF00216;
00328 00222          IEND;
00329 00223          DF00226 = SQT(DF00224);
00330 00224          IF DF00222 == 0;
00331 00225              DF00222 = 1;
00332          IEND;
00333 00226          IF DF00226 == 0;
00334 00227              DF00226 = 1;
00335 00228          IEND;
00336 00229          IF DF00226 == 0;
00337 00230              DF00226 = 1;
00338 00231          IEND;
00339 00232          DF00228 = ML01124 * DF00222 / DF00226 * 60;
00340 00233          DL00230 = DF00228;

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00341
00342 00234      DL00012 = ML10800[IDX_CX];
00343 00235      DF00020 = DL00012 / 60 / ML01146 * 1000;
00344 00236      DF00022 = DL00012 / 60 / ML01148 * 1000;
00345 00237      DL00014 = DF00020;
00346 00238      DL00016 = DF00022;
00347
00348 00239      IF DL00014 > 32767;
00349 00240          DL00014 = 32767;
00350 00241      IEND;
00351 00242      IF DL00016 > 32767;
00352 00243          DL00016 = 32767;
00353 00244      IEND;
00354
00355      " 1st Cutter X Speed(UPCX) < 1구간 Cutter X Speed(LOCX)
00356 00245      IF ML15292 < ML15294
00357 00246          Scribe_Speed_1st = ML15294 * 1.414 * 60;
00358 00247      ELSE;
00359 00248          Scribe_Speed_1st = ML15292 * 1.414 * 6
00360 00249      IEND;
00361
00362 00250      Scribe_Speed_2nd = DL00230;
00363
00364      " 3rd Cutter X Speed(UPCX) < 3rd Cutter X Speed(LOCX)
00365 00251      IF ML15296 < ML15298;
00366 00252          Scribe_Speed_3rd = ML15298 * 1.414 * 60;
00367 00253      ELSE;
00368 00254          Scribe_Speed_3rd = ML15296 * 1.414 * 6
00369 00255      IEND;
00370
00371 00256      DL00030 = Scribe_Speed_1st;
00372 00257      DL00032 = Scribe_Speed_2nd;
00373 00258      DL00034 = Scribe_Speed_3rd;
00374
00375 00259      PFORK Label11 Label12 Label13
00376      Label11:
00377
00378 00260          IF (DB00020 & !DB00021 & !DB00022) ==          " Recipe Data
00379 00261              ML04850 = ML43412 * 2 * 60;
00380
00381 00262              IOW DB000000 == 1          " 1st Cutting Cmt
00382
00383              " 2nd
00384 00263              MW15520 = 7;          " Torque Setting
00385 00264              MW15525 = 9;          " Axis Select
00386 00265              MSEE MPS253;          " CZ Torque Set
00387
00388 00266              ML16008 = ML15226          " [UPCZ] Upper Cutter Z Moving Position
00389      Write
00390 00267              IOW MB045778 == 1          " Cutter Z Move Permission
00391
00392 00268              MVS [CZ]ML16008 FML04850;
00393
00394 00269              IF DB000011 == 0          " 3rd Set Flag (0:Use 1:NotUse)
00395 00270                  IOW DB000001 == 1          " 2nd Cutting Cmt
00396
00397              " 3rd
00398 00271              MW15520 = 11;          " Torque Setting
00399 00272              MW15525 = 9;          " Axis Select
00400 00273              MSEE MPS253;          " CZ Torque Set
00401
00402              " 3rd Cutter Z Scribe Infeed Volume(um)
00403              " Upper Head
00404
00405              " UPCZ, UPRZ
00406 00274              ML15274 = ILA116 + GL100062[SC_PRT_Addr_L];
00407
00408 00275              ML16008 = ML15274          " [CZ] Upper Cutter Z Moving Position Wr
00409      ite
00410 00276              IOW MB045778 == 1          " Cutter Z Move Permission
00411
00412 00277              MVS [CZ]ML16008 FML04850;
00413 00278              IEND;
00414
00415 00279              IOW DB000002 == 1          " 3rd Cutting Cm
00416
00417 00280              MSEE MPS255;          " CZ Torque Reset
00418 00281              IEND;
00419
00420 00282              JOINTO Label1X;
00421
00422      Label12:
00423
00424 00283              FMX tDL00012;
00425 00284              IAC tDL00014;

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00426 00285      IDC tDL00016;
00427
00428 00286      ACCMODE M4;
00429
00430 00287      S{MB047351} = DB00020          " Scribe Run Check
00431
00432      " 1st Cutting(Not) / 3rd Cutting(Not)
00433 00288      IF DB000010 & DB000011 == 1;
00434 00289          DB000000 = 1;
00435
00436 00290          IF DB000031 == 1          " 2nd Cutting Enable
00437 00291              MVS [CX]DL00510 [CS]DL00314 FDL00032 FE0;
00438 00292          IEND;
00439
00440 00293          IOW IBA00C1 == 1;
00441 00294          DB000001 = 1;
00442 00295          DB000002 = 1;
00443 00296      ELSE;
00444          " 1st Cutting(Not) / 3rd Cutting(Use)
00445 00297          IF DB000010 == 1;
00446 00298              DB000000 = 1;
00447
00448 00299          IF DB000031 == 1          " 2nd Cutting Enable
00449 00300              MVS [CX]DL00510 [CS]DL00314 FDL00032 FEDL00034;
00450 00301          IEND;
00451
00452 00302          DB000001 = 1;
00453
00454 00303          IF DB000032 == 1          " 3rd Cutting Enable
00455 00304              MVS [CX]DL00520 [CS]DL00324 FDL00034 FE0;
00456 00305          IEND;
00457
00458 00306          DB000002 = 1;
00459 00307      ELSE;
00460          " 1st Cutting(Use) / 3rd Cutting(Not)
00461 00308          IF DB000011 == 1
00462              IF DB000030 == 1          " 1st Cutting Enable
00463 00309                  MVS [CX]DL00500 [CS]DL00304 FDL00030 FEDL00030;
00464 00310                  IEND;
00465 00311
00466                  DB000000 = 1;
00467 00312
00468                  IF DB000031 == 1          " 2nd Cutting Enable
00469 00313                      MVS [CX]DL00510 [CS]DL00314 FDL00032 FE0;
00470 00314                      IEND;
00471 00315
00472                      DB000001 = 1;
00473 00316                      DB000002 = 1;
00474 00317
00475 00318                      ELSE;
00476                          " 1st Cutting(Use) / 3rd Cutting(Use)
00477 00319                          IF DB000030 == 1          " 1st Cutting Enable
00478 00320                              MVS [CX]DL00500 [CS]DL00304 FDL00030 FEDL00030;
00479 00321                              IEND;
00480
00481 00322                              DB000000 = 1;
00482
00483 00323                              IF DB000031 == 1          " 2nd Cutting Enable
00484 00324                                  MVS [CX]DL00510 [CS]DL00314 FDL00032 FEDL00034;
00485 00325                                  IEND;
00486
00487 00326                                  DB000001 = 1;
00488
00489 00327                                  IF DB000032 == 1          " 3rd Cutting Enable
00490 00328                                      MVS [CX]DL00520 [CS]DL00324 FDL00034 FE0;
00491 00329                                      IEND;
00492
00493 00330                                      DB000002 = 1;
00494 00331                                      IEND;
00495 00332                                  IEND;
00496 00333                                  IEND;
00497
00498 00334      MB047351 = 0;          " Scribe Run Chec
00499 00335      MB047350 = 0;          " Scribe_StopRetr

00500
00501 00336      ACCMODE M0;
00502
00503 00337      JOINTO LabelX;
00504
00505      Label3:
00506
00507      " UCAMX Escape Position
00508 00338      ML16066 = ML43330          " UCAMX Waiting Position
00509 00339      IOW MB04577A == 1          " UCAMX Moving Permission
00510 00340      VEL [CAMX]ML43312;
00511 00341      MOV [CAMX]ML16066;

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00512 00342      IOW MB046010 == 1      " UCAMX Waiting Position Chec
00513
00514 00343      JOINTO LabelX;
00515 00344      LabelX: PJOINT;
00516
00517 00345      MB15420F = 0;          " Scribe Executing Flag
00518
00519 00346      RET;
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